

Protocol $\Pi_{\text{Slot}}^{\mathcal{F}_{\text{AuthField-Ring-Bool}}}$

Setup state. Fix the MLDSA parameters, matrix A , message representative μ , slot count N_s , and agreed circuits $C_{\text{pre}}, C_{\text{post}}$. Key generation supplies $\llbracket \mathbf{s} \rrbracket_D^{\text{BDOZ}}, \llbracket \mathbf{e} \rrbracket_D^{\text{BDOZ}}$ with coefficients in $[-\eta, \eta]$ and $\llbracket \mathbf{s} \rrbracket_q^{\text{BDOZ}}$, all bound to the same public key and common $\mathbf{t}, \mathbf{t}_0$. Let $d_{\text{all}} = \langle P_1, \dots, P_n \rangle$. P_1 evaluates C_{post} ; $P_2, \dots, P_n$ garble it. Every P_p holds a global key $\Delta_p \in \{0, 1\}^{128}$.

Representations. $\llbracket x \rrbracket_2^{\text{WRK}}$ is the WRK encoding of a wire carrying the bit x : all parties hold $\llbracket \lambda_x \rrbracket_2^{\text{BDOZ}}$; each garbler P_p holds a zero label $L_p^0(x)$; P_1 holds $t_x = x \oplus \lambda_x$ and $L_p(x) = L_p^0(x) \oplus t_x \Delta_p$. For bit vectors, bitwise.

Before the challenge.

1. All parties invoke the following four commands. Each returned authenticated vector pair is shown on the left:

$$\begin{aligned} (\llbracket \mathbf{R}_y \rrbracket_q^{\text{BDOZ}}, \llbracket \text{bin}_a(\mathbf{R}_y) \rrbracket_2^{\text{BDOZ}}) &\leftarrow \text{eDaBit}(q, a, n_z), \\ (\llbracket \mathbf{R}_w \rrbracket_q^{\text{BDOZ}}, \llbracket \text{bin}_{23}(\mathbf{R}_w) \rrbracket_2^{\text{BDOZ}}) &\leftarrow \text{eDaBit}(q, 23, n_w), \\ (\llbracket \mathbf{R}_H \rrbracket_D^{\text{BDOZ}}, \llbracket \text{bin}_\nu(\mathbf{R}_H) \rrbracket_2^{\text{BDOZ}}) &\leftarrow \text{eDaBit2k}(\nu, m), \\ (\llbracket \boldsymbol{\rho} \rrbracket_q^{\text{BDOZ}}, \llbracket \text{bin}_{23}(\boldsymbol{\rho}) \rrbracket_2^{\text{BDOZ}}) &\leftarrow \text{eDaBit}(q, 23, n_z), \end{aligned}$$

and $\llbracket \mathbf{e}_w \rrbracket_q^{\text{BDOZ}}$ with coefficients uniform in $[-\eta, \eta]$ (dealt directly; no Boolean half of $\mathbf{e}_w$ is used). The fourth supplies the response pad $\boldsymbol{\rho}$, used for one challenge only.

2. Locally compute

$$\begin{aligned} \llbracket \mathbf{y} \rrbracket_q^{\text{BDOZ}} &= \gamma_1 \mathbf{1} - \llbracket \mathbf{R}_y \rrbracket_q^{\text{BDOZ}}, \\ \llbracket \mathbf{w} \rrbracket_q^{\text{BDOZ}} &= A \llbracket \mathbf{y} \rrbracket_q^{\text{BDOZ}} + \llbracket \mathbf{e}_w \rrbracket_q^{\text{BDOZ}}, \\ \llbracket \boldsymbol{\delta}_w \rrbracket_q^{\text{BDOZ}} &= \llbracket \mathbf{w} \rrbracket_q^{\text{BDOZ}} + \llbracket \mathbf{R}_w \rrbracket_q^{\text{BDOZ}}. \end{aligned}$$

Invoke $(\text{Open}, \llbracket \boldsymbol{\delta}_w \rrbracket_q^{\text{BDOZ}}, d_{\text{all}})$. (The implementation pads with $+\mathbf{R}_w$; the figure pads with $-\mathbf{R}_w$.)

3. Invoke BoolEval on C_{pre} , using $\llbracket \text{bin}_a(\mathbf{R}_y) \rrbracket_2^{\text{BDOZ}}, \llbracket \text{bin}_{23}(\mathbf{R}_w) \rrbracket_2^{\text{BDOZ}}, \llbracket \text{bin}_\nu(\mathbf{R}_H) \rrbracket_2^{\text{BDOZ}}$ as inputs and the coefficients of $\boldsymbol{\delta}_w$ as public constants. All operations below are applied coefficientwise:

$$\begin{aligned} \mathbf{w} &= (\boldsymbol{\delta}_w - \mathbf{R}_w) \bmod q, \\ (\mathbf{w}_1, \mathbf{w}_0) &= \text{Decompose}(\mathbf{w}, 2\gamma_2), \\ \mathbf{M} &= \mathbf{R}_H + (\text{shift}(\mathbf{R}_y, 2\gamma_1) \parallel \text{shift}(\gamma_2 \mathbf{1} - \mathbf{w}_0, 2\gamma_2)) \pmod{D}, \\ \partial_i &= [U_i \leq 2\beta] \vee [U_i \geq S_i - 2\beta], \\ \text{cnt}_i &= \sum_{k < i} \partial_k, \\ u_{j,i} &= \partial_i \wedge [\text{cnt}_i = j], \\ \widehat{M}_j &= \left(\bigoplus_i u_{j,i} M_i \right) \oplus \left(L \cdot [\sum_i u_{j,i} = 0] \right), \\ \text{ovf} &= [\text{cnt}_m > N_s]. \end{aligned}$$

Here $U = (\mathbf{R}_y \parallel \gamma_2 \mathbf{1} - \mathbf{w}_0)$ and $\mathbf{S}$ carry the arguments of shift , the slot index runs over $0 \leq j < N_s$, and $\mathbf{u}_j = (u_{j,i})_{i < m}$ is one-hot, all zero for an unused slot. The selection $\widehat{M}_j$ is realised by a stable compaction network of $m\nu \lceil \log_2 m \rceil$ AND gates (202,752 for MLDSA44), not by the literal $N_s m \nu$ -gate selection circuit. Return only $\llbracket \mathbf{w}_1 \rrbracket_2^{\text{BDOZ}}, \{\llbracket \mathbf{u}_j \rrbracket_2^{\text{BDOZ}}, \llbracket \widehat{M}_j \rrbracket_2^{\text{BDOZ}}\}_{j < N_s}, \llbracket \text{ovf} \rrbracket_2^{\text{BDOZ}}$, and invoke $(\text{Open}, (\llbracket \mathbf{w}_1 \rrbracket_2^{\text{BDOZ}}, \llbracket \text{ovf} \rrbracket_2^{\text{BDOZ}}), d_{\text{all}})$. If $\text{ovf} = 1$, retire the execution and restart from step 1.

Conversion $\llbracket x \rrbracket_2^{\text{BDOZ}} \rightarrow \llbracket x \rrbracket_2^{\text{WRK}}$ (still before the challenge). For every bit x of $\text{bin}_\nu(\widehat{M}_j)$, $\text{bin}_{23}(\rho)$ and $\mathbf{u}_j$ ($j < N_s$), which are the fixed inputs of C_{post} :

$$\begin{aligned} \llbracket \lambda_x \rrbracket_2^{\text{BDOZ}} &\leftarrow \text{AuthRandomBit}, & t_x &= \text{Open}(\llbracket x \rrbracket_2^{\text{BDOZ}} \oplus \llbracket \lambda_x \rrbracket_2^{\text{BDOZ}}, d_{\text{all}}), \\ P_p &\rightarrow P_1 : L_p(x) = L_p^0(x) \oplus t_x \Delta_p. \end{aligned}$$

All t_x are opened in one checked opening. Then garble C_{post} (WRK) offline; its late input wires $\widehat{\delta}_j[\ell]$ get fresh masks that are never opened, and its output masks are opened to P_1 . The $\mathbf{u}_j$ wires are read by no gate of C_{post} ; they exist only to be XOR-ed in step 6. $\llbracket \mathbf{u}_j \rrbracket_2^{\text{BDOZ}}, \llbracket \widehat{M}_j \rrbracket_2^{\text{BDOZ}}, \llbracket \text{bin}_{23}(\rho) \rrbracket_2^{\text{BDOZ}}$ are then discarded.

Retained state. Retain $\{\llbracket \mathbf{u}_j \rrbracket_2^{\text{WRK}}, \llbracket \widehat{M}_j \rrbracket_2^{\text{WRK}}\}_{j < N_s}, \llbracket \text{bin}_{23}(\rho) \rrbracket_2^{\text{WRK}}$, the garbled C_{post} , $\llbracket \mathbf{R}_H \rrbracket_D^{\text{BDOZ}}, \llbracket \mathbf{s} \rrbracket_D^{\text{BDOZ}}, \llbracket \mathbf{e} \rrbracket_D^{\text{BDOZ}}, \llbracket \mathbf{s} \rrbracket_q^{\text{BDOZ}}, \llbracket \mathbf{y} \rrbracket_q^{\text{BDOZ}}, \llbracket \rho \rrbracket_q^{\text{BDOZ}}$, and $\mathbf{w}_1$.

After the challenge.

4. **Challenge.** Compute the parameter-set-specific $\tilde{c} = \text{H}(\mu \| \text{w1Encode}(\mathbf{w}_1))$ and $c = \text{SampleInBall}(\tilde{c})$. Bind both to this attempt and the following affine expressions.
5. **Flight 1: checked masked openings.** Over $R_D^{\ell+k}$ and R_q^ℓ respectively, locally compute

$$\begin{aligned} \llbracket \delta_H \rrbracket_D^{\text{BDOZ}} &= (\beta \mathbf{1} - c \llbracket \mathbf{s} \rrbracket_D^{\text{BDOZ}}, \beta \mathbf{1} + c \llbracket \mathbf{e} \rrbracket_D^{\text{BDOZ}}) - \llbracket \mathbf{R}_H \rrbracket_D^{\text{BDOZ}}, \\ \llbracket \bar{\mathbf{V}} \rrbracket_q^{\text{BDOZ}} &= (\gamma_1 + \beta) \mathbf{1} - \llbracket \mathbf{y} \rrbracket_q^{\text{BDOZ}} - c \llbracket \mathbf{s} \rrbracket_q^{\text{BDOZ}} + \llbracket \rho \rrbracket_q^{\text{BDOZ}}. \end{aligned}$$

Invoke $(\text{Open}, (\llbracket \delta_H \rrbracket_D^{\text{BDOZ}}, \llbracket \bar{\mathbf{V}} \rrbracket_q^{\text{BDOZ}}), d_{\text{all}})$ once for all $(\ell + k)N$ and ℓN coefficients.

6. **Flight 2: routing and local evaluation.** The checked residues δ_H fix the routing. For $0 \leq j < N_s$ and $0 \leq \ell < \nu$, P_1 forms by free XOR of the retained encodings, without communication,

$$\llbracket \widehat{\delta}_j[\ell] \rrbracket_2^{\text{WRK}} = \bigoplus_{i: \delta_{H,i}[\ell]=1} \llbracket u_{j,i} \rrbracket_2^{\text{WRK}}.$$

Write d for the wire carrying this XOR-ed encoding and f for the input wire of the pre-garbled C_{post} that $\widehat{\delta}_j[\ell]$ enters. Both carry the same bit, but d is under $\lambda_d = \bigoplus \lambda_{u_{j,i}}$, $L_p^0(d) = \bigoplus L_p^0(u_{j,i})$, while f is under its own λ_f , $L_p^0(f)$ sampled at garbling time. Solder d to f : the garblers reveal to P_1 , authenticated, $\varepsilon = \lambda_d \oplus \lambda_f$ and $L_p^0(d) \oplus L_p^0(f) \oplus \varepsilon \Delta_p$ (one 1+16n-byte block per wire per garbler, sent from its share of ε , MACs and keys), and P_1 swaps mask bit and labels:

$$t_f = t_d \oplus \varepsilon, \quad L_p(f) = L_p(d) \oplus L_p^0(d) \oplus L_p^0(f) \oplus \varepsilon \Delta_p.$$

P_1 then evaluates C_{post} :

$$\begin{aligned} \llbracket K_j \rrbracket_2^{\text{WRK}} &= (\llbracket \widehat{M}_j \rrbracket_2^{\text{WRK}} + \llbracket \widehat{\delta}_j \rrbracket_2^{\text{WRK}}) \bmod D, \\ \llbracket r \rrbracket_2^{\text{WRK}} &= \bigwedge_{j=0}^{N_s-1} \llbracket K_j[b] \rrbracket_2^{\text{WRK}}, \\ \llbracket \mathbf{Z}_{\text{out}} \rrbracket_2^{\text{WRK}} &= \llbracket r \rrbracket_2^{\text{WRK}} \cdot \llbracket \text{bin}_{23}(\rho) \rrbracket_2^{\text{WRK}}, \end{aligned}$$

and decodes $r, \mathbf{Z}_{\text{out}}$ locally with the output masks opened to it before the challenge. It obtains no individual predicate or rejected response.

7. **Completion.** If $r = 1$, the evaluator recovers $\mathbf{z} = (\gamma_1 + \beta)\mathbf{1} - ((\bar{\mathbf{V}} - \mathbf{Z}_{\text{out}}) \bmod q)$ and locally computes:

$$\mathbf{d}_0 = \mathbf{c}t_0, \quad \mathbf{h} = \text{MakeHint}_q(-\mathbf{d}_0, A\mathbf{z} - \mathbf{c}t + \mathbf{d}_0, 2\gamma_2).$$

Reject if $\|\mathbf{d}_0\|_\infty \geq \gamma_2$, $\text{HW}(\mathbf{h}) > \omega$, or public MLDSA verification fails; otherwise return $(\tilde{\mathbf{c}}, \mathbf{z}, \mathbf{h})$.
Retire the backend execution after this decision.

Figure 1: The slot-restricted signing attempt. Only steps 3, 5 and 6 differ from the two-round protocol.

The idea. By the three cases of shift in Equation (3), a coefficient with $2\beta < U_i < S_i - 2\beta$ accepts for every reachable H , so its online test is redundant; only the $\partial_i = 1$ coefficients can fail, and in expectation there are fewer than 3.3 of them out of $m \geq 2048$. Step 3 compacts these into N_s slots fixed before the challenge. Two things keep the attempt at two flights. First, the routing $\mathbf{u}_j \mapsto \hat{\delta}_j$ is $\mathbb{F}_2$ -linear with coefficients that become public only at flight 1, so the evaluator forms it by XOR-ing active labels it already holds: the XOR routing itself needs no AND gate and no garbling material, and there is no **PublicBits** input. Second, the response no longer rides on the boundary sum through base; it travels through the field one-time pad $\boldsymbol{\rho}$, whose masked value $\bar{\mathbf{V}}$ is opened in flight 1 and whose release the circuit gates on r . Phase 2 therefore carries $N_s b$ masked-sum gates in place of mb , and no response additions at all.

Choosing N_s . Write $m_\partial = \text{cnt}_m$. Since U_i is uniform on $2\gamma_1$ values for a response coefficient and on $2\gamma_2 + 1$ values for a low-bit coefficient,

$$p_z = \frac{4\beta + 1}{2\gamma_1}, \quad p_w = \frac{4\beta + 2}{2\gamma_2 + 1}, \quad \lambda = \mathbb{E}[m_\partial] = n_z p_z + n_w p_w,$$

and m_∂ is a sum of m independent indicators under the usual heuristic that the $\mathbf{w}_0$ coefficients are uniform and independent. The multiplicative Chernoff bound with $t = N_s + 1$ gives

$$\Pr[m_\partial > N_s] \leq \left(\frac{e\lambda}{t}\right)^t e^{-\lambda},$$

and the exact convolution $\text{Bin}(n_z, p_z) * \text{Bin}(n_w, p_w)$ is only a few bits tighter.

	MLDSA44	MLDSA65	MLDSA87
λ	2.9108	3.2632	2.7067
Chernoff, $N_s = 64$	$2^{-201.7}$	$2^{-191.5}$	$2^{-208.2}$
Exact, $N_s = 64$	$2^{-207.3}$	$2^{-196.8}$	$2^{-213.3}$

Why the target is 2^{-196} and not 2^{-128} . Overflow discards the candidate before the challenge, but the surviving output distribution is $\Pr[\mathbf{z} = \mathbf{z}_0] \propto \mathbf{1}[\mathbf{z}_0 \in B] \cdot \mathbf{1}[m_\partial(\mathbf{z}_0 - c\mathbf{s}) \leq N_s]$, which depends on $c\mathbf{s}$. The per-attempt statistical distance is at most $\Pr[\text{ovf}]/\Pr[\text{acc}] \leq 2^{2.4} \Pr[\text{ovf}]$, and a key signing 2^{64} messages makes $Q \approx 2^{66}$ attempts, so a lifetime bound of 2^{-128} requires $\Pr[\text{ovf}] \leq 2^{-196}$. We instantiate $N_s = 64$, which the exact tail certifies for all three parameter sets, with MLDSA65 the binding case at 0.8 bits of margin. Certifying the same claim from the Chernoff bound alone needs $N_s = 68$ there; $N_s = 48$ supports only a per-attempt claim.